immediate explanation. Even though electrons couldn't linger in between the allowed orbits, they could *jump* from one to another. An electron could fall from a higher-energy orbit to a lower-energy one by emitting light with just the right energy to compensate, or it could leap upward in energy by absorbing an appropriate amount of energy from ambient light. Because the orbits themselves were quantized, we should only see specific energies of light interacting with the electrons. Together with Planck's idea that the frequency of light is related to its energy, this explained why physicists saw only certain frequencies being emitted or absorbed.

By comparing his predictions to the observed emission of light by hydrogen, Bohr was able to not simply posit that only some electron orbits were allowed, but calculate which ones they were. Any orbiting particle has a quantity called the *angular momentum*, which is easy to calculate—it's just the mass of the particle, times its velocity, times its distance from the center of the orbit. Bohr proposed that an allowed electron orbit was one whose angular momentum was a multiple of a particular fundamental constant. And when he compared the energy that electrons should emit when jumping between orbits to what was actually seen in light emitted from hydrogen gas, he could figure out what that constant needed to be in order to fit the data. The answer was Planck's constant, h. Or more specifically, the modified h-bar version, $\hbar = h/2\pi$.

That's the kind of thing that makes you think you're on the right track. Bohr was trying to account for the behavior of electrons in atoms, and he posited an ad hoc rule according to which they could only move along certain quantized orbits, and in order to fit the data his rule ended up requiring a new constant of nature, and that new constant was the same as the new constant that Planck was forced to invent when he was trying to account for the behavior of photons. All of this might seem ramshackle and a bit sketchy, but taken together it appeared as if something profound was happening in the realm of atoms and particles, something that didn't fit comfortably with the sacred rules of classical mechanics. The ideas of this period are now sometimes described under the rubric of "the old quantum theory," as opposed to "the new quantum theory" of Heisenberg and Schrödinger that came along in the late 1920s.